

EMILIA Protocol

Evidence and durable-consequence control for agentic AI · proposed for AAIF TC review ·

non-binding feedback sought

MCP governs how an agent **connects**. Agent frameworks govern how it **executes**. Policy systems govern what it **may call**. EMILIA answers the consequence question: when an action moves money, changes infrastructure, deletes data, or creates irreversible state, what proves **who authorized the exact material action** - and what prevents an ambiguous provider timeout from becoming a blind replay?

AgentROA what an agent may call	ORPRG policy permitted the effect	EMILIA who authorized the exact action, plus durable consequence control
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Decision logs are testimony. Verifiable receipts plus executor-owned state are evidence.

WHAT IT IS

An open verification substrate plus a deny-by-default Gate. Native receipts are verified under relying-party-pinned trust, matched to one material action, and evaluated against an explicit evidence requirement. Gate separately authorizes, reserves one-time or bounded state before invocation, records `EXECUTED` or `INDETERMINATE`, refuses blind retry, and reconciles only authenticated provider evidence.

IT IS REAL - AND THE CLAIMS ARE BOUNDED

- **Active individual Internet-Drafts**, including authorization receipts, quorum, AEC, action evidence graph, and enforcement-point work. These are author submissions, not IETF-adopted standards or RFCs.
- **Reference verification** in JavaScript, Python, and Go: 18 suites / 251 current vectors. These are same-team ports, so agreement is cross-language consistency - not three independent implementations.
- **Formal gates**: 26 TLA+ safety properties and 35 Alloy facts / 32 assertions across four CI-run models, with zero reported errors or counterexamples in the declared scopes.
- **External evidence, precisely scoped**: an externally authored Rust verifier is rebuilt from pinned public source and tested against a pinned 16-suite / 164-vector bundle plus 359 hostile cases. Strict independently attested construction acceptance remains pending.
- **Runnable consequence demo**: provider executes but times out; Gate records indeterminate, refuses a second effect, and reconciles authenticated provider evidence.

Private architecture drafts: CAID and Bounded Capability are prepared EMILIA drafts, not posted standards. AEC is an active public individual Internet-Draft. None is described as IETF-adopted.

PROPOSAL FACTS (FOR THE TC)

License: Apache-2.0	Governance: public repository, DCO-gated contributions
Repo: github.com/emiliaprotocol/emilia-protocol	Adoption: published npm/PyPI/Go packages
Verification: offline; no EMILIA backend required	Standards status: active individual I-Ds, no IETF endorsement claimed
Composes with: MCP · AgentROA · ORPRG · WIMSE	Commercial layer: executor-side Consequence Firewall

WHAT WE ARE ASKING

Early, non-binding Technical-Committee feedback on fit, terminology, and interoperability. EMILIA is not asking the TC to adopt a private roadmap or to treat an individual draft as a standard. The useful question is whether an open, executor-side evidence and consequence layer belongs beside the agent runtime and policy components already forming the stack.

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Verification establishes the stated cryptographic, matching, and lifecycle properties - not civil identity, legal validity, business wisdom, physical outcome, or regulatory compliance.